

Circuit Parameters

$R[0][0] = 1 \text{ k}\Omega$ (ON weight)

$R[0][1] = 2 \text{ k}\Omega$ (OFF weight)

$R[1][0] = 2 \text{ k}\Omega$ (OFF weight)

$R[1][1] = 1 \text{ k}\Omega$ (ON weight)

Task 1

Setup: $V_{\text{row0}} = 1 \text{ V}$, $V_{\text{col0}} = 0 \text{ V}$ (virtual ground), $V_{\text{row1}} = 0 \text{ V}$, $V_{\text{col1}} = 0 \text{ V}$.

In the ideal case all unselected lines are grounded so no sneak paths exist. Only $R[0][0]$ carries current:

$$\begin{aligned} I_{\text{col0}} &= (V_{\text{row0}} - V_{\text{col0}}) / R[0][0] \\ &= (1 \text{ V} - 0 \text{ V}) / 1 \text{ k}\Omega \\ &= 1.0 \text{ mA} \end{aligned}$$

Ideal $I_{\text{col0}} = 1.0 \text{ mA}$

Task 2

Setup: $V_{\text{row0}} = 1 \text{ V}$, $V_{\text{col0}} = 0 \text{ V}$. Row 1 and Col 1 are floating (undriven).

Let V_{col1} and V_{row1} be the unknown floating node voltages.

KCL at node V_{col1} (sum of currents leaving = 0)

Current from Row 0 into Col 1 via $R[0][1]$, and current from Col 1 into Row 1 via $R[1][1]$:

$$\begin{aligned} (V_{\text{col1}} - V_{\text{row0}}) / R[0][1] + (V_{\text{col1}} - V_{\text{row1}}) / R[1][1] &= 0 \\ (V_{\text{col1}} - 1) / 2000 + (V_{\text{col1}} - V_{\text{row1}}) / 1000 &= 0 \quad \dots (1) \end{aligned}$$

KCL at node V_{row1} (sum of currents leaving = 0)

Current from Row 1 into Col 1 via $R[1][1]$, and current from Row 1 into Col 0 via $R[1][0]$:

$$\begin{aligned} (V_{\text{row1}} - V_{\text{col1}}) / R[1][1] + (V_{\text{row1}} - V_{\text{col0}}) / R[1][0] &= 0 \\ (V_{\text{row1}} - V_{\text{col1}}) / 1000 + V_{\text{row1}} / 2000 &= 0 \quad \dots (2) \end{aligned}$$

Solving equation (2) for V_{row1} :

$$\begin{aligned} 2(V_{\text{row1}} - V_{\text{col1}}) + V_{\text{row1}} &= 0 \\ 2 \cdot V_{\text{row1}} - 2 \cdot V_{\text{col1}} + V_{\text{row1}} &= 0 \\ 3 \cdot V_{\text{row1}} &= 2 \cdot V_{\text{col1}} \\ V_{\text{row1}} &= (2/3) \cdot V_{\text{col1}} \quad \dots (3) \end{aligned}$$

Substituting (3) into equation (1):

$$\begin{aligned} (V_{\text{col1}} - 1) / 2 + (V_{\text{col1}} - (2/3) \cdot V_{\text{col1}}) / 1 &= 0 \quad (\text{in k}\Omega \text{ units}) \\ (V_{\text{col1}} - 1) / 2 + (1/3) \cdot V_{\text{col1}} &= 0 \\ V_{\text{col1}} / 2 - 1/2 + V_{\text{col1}} / 3 &= 0 \\ (3 \cdot V_{\text{col1}} + 2 \cdot V_{\text{col1}}) / 6 &= 1/2 \\ 5 \cdot V_{\text{col1}} / 6 &= 1/2 \\ V_{\text{col1}} &= 3/5 = 0.6 \text{ V} \end{aligned}$$

$$V_{\text{col1}} = 0.6 \text{ V}$$

$$V_{\text{row1}} = (2/3) \times 0.6 = 0.4 \text{ V}$$

(c) Actual I_{col0} Including Sneak Path

Two paths contribute current to Col 0:

Path 1 — Direct path: Row 0 $\rightarrow$ $R[0][0]$ $\rightarrow$ Col 0

$$\begin{aligned} I_{direct} &= (V_{row0} - V_{col0}) / R[0][0] \\ &= (1 \text{ V} - 0 \text{ V}) / 1 \text{ k}\Omega \\ &= 1.0 \text{ mA} \end{aligned}$$

Path 2 — Sneak path: Row 1 $\rightarrow$ $R[1][0]$ $\rightarrow$ Col 0

$$\begin{aligned} I_{sneak} &= (V_{row1} - V_{col0}) / R[1][0] \\ &= (0.4 \text{ V} - 0 \text{ V}) / 2 \text{ k}\Omega \\ &= 0.2 \text{ mA} \end{aligned}$$

Total measured current:

$$\begin{aligned} I_{col0} &= I_{direct} + I_{sneak} \\ &= 1.0 \text{ mA} + 0.2 \text{ mA} \\ &= 1.2 \text{ mA} \end{aligned}$$

$$\text{Read error} = (1.2 - 1.0) / 1.0 \times 100\% = 20\% \text{ error}$$

Task 3

How Sneak Paths Corrupt MVM Results

When row 1 and col 1 are left floating, current from row 0 leaks through $R[0][1]$ into col 1, travels down through $R[1][1]$ into row 1, and returns through $R[1][0]$ to col 0 — adding 0.2 mA of parasitic current that was never intended. This sneak path inflates the sensed current from 1.0 mA to 1.2 mA, introducing a 20% read error that directly corrupts the Matrix-Vector Multiplication result since col 0 now reports a weighted sum that includes contributions from unselected cells. In large crossbar arrays this problem compounds significantly because every additional row introduces more sneak paths, making accurate analog MVM computation impossible without access devices such as transistors (1T1R) or selector diodes to isolate unselected cells.